

# INDUSTRIAL INTERNSHIP REPORT

## Automotive E-Commerce & Full-Stack Information System

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**Domain:** Full Stack Web Development

**Industry Partner:** UniConverge Technologies Pvt Ltd (UCT)

**Execution Partner:** upSkill Campus (USC) & The IoT Academy

**Submission Date:** August 2026

### EXECUTIVE SUMMARY

This report provides details of the Industrial Internship provided by **upSkill Campus** and **The IoT Academy** in collaboration with Industrial Partner **UniConverge Technologies Pvt Ltd (UCT)**. This internship was focused on a project and problem statement provided by UCT to be finished, including documentation, within a 6-week timeframe.

My project was centered on an **Automotive E-Commerce Marketplace & Core Information System**. The objective was to engineer an intuitive, resilient platform supporting catalog exploration across diverse part categories (Brakes, Electrical, Fluids, Lighting), persistent multi-item carts, dynamic pricing algorithms, and transactional data persistence. This internship provided practical exposure to industrial development standards, architectural designs, and real-world system testing, proving to be an exceptional professional learning experience.

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## 1. PREFACE

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Engaging in an industry-backed internship is vital for building competency beyond classroom theory. Over the 6-week program, work progressed across structured phases: exploring the problem statement and learning about UCT's industrial solutions, following technical guidelines to architect a solution, core coding and state handling, iterative refinement, system performance validation, and final report submission[span\_10](start\_span)[span\_10](end\_span).

I would like to express my sincere gratitude to the technical mentors at UniConverge Technologies, upSkill Campus, and The IoT Academy for their constant support and structured learning modules[span\_11](start\_span)[span\_11](end\_span). My message to my peers and juniors is to emphasize writing modular code, mastering underlying architectural principles such as OOP and database query lifecycles, and focusing on solving tangible end-user problems[span\_12](start\_span)[span\_12](end\_span)[span\_13](start\_span)[span\_13](end\_span)[span\_14](start\_span)[span\_14](end\_span)[span\_15](start\_span)[span\_15](end\_span).

## 2. INTRODUCTION

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### 2.1 About UniConverge Technologies Pvt Ltd (UCT)

Established in 2013, UniConverge Technologies Pvt Ltd (UCT) works in the Digital Transformation domain, delivering industrial solutions with a core focus on sustainability and Return on Investment (RoI)[span\_16](start\_span)[span\_16](end\_span). UCT leverages cutting-edge technologies including Internet of Things (IoT), Cybersecurity, Cloud Computing (AWS, Azure), Machine Learning, Java Full Stack, Python, and Communication Technologies (4G/5G/LoRaWAN)[span\_17](start\_span)[span\_17](end\_span). Flagship industrial platforms developed by UCT include:

- **UCT Insight:** An enterprise IoT platform designed for rapid deployment of IoT applications providing operational insights(start\_span)(end\_span). Built using Java for the backend and ReactJS for the frontend, with support for MySQL and NoSQL databases and standard protocols like MQTT, CoAP, and HTTP(start\_span)(end\_span).
- **Factory Watch:** A smart factory SaaS platform providing Overall Equipment Effectiveness (OEE), predictive maintenance, machine condition monitoring, and digital twins for industrial assets(start\_span)(end\_span).
- **LoRaWAN Solutions & Predictive Maintenance:** Smart city, metering, and industrial health monitoring systems calculating the Remaining Useful Life (RUL) of production machinery(start\_span)(end\_span).

## 2.2 About upSkill Campus (USC) & The IoT Academy

**upSkill Campus (USC):** A career development platform delivering accessible, scalable coaching and practical upskilling opportunities aimed at training 1 million learners(start\_span)(end\_span).

**The IoT Academy:** The EdTech division of UCT running executive certification programs in collaboration with leading institutions (EICT Academy, IIT Kanpur, IIT Roorkee, and IIT Guwahati) across core software and engineering domains(start\_span)(end\_span).

## 2.3 Objectives of this Internship Program

- Gain practical hands-on experience working on industrial problem statements(start\_span)(end\_span).
- Develop software solutions that solve real-world user and business challenges(start\_span)(end\_span).
- Improve technical proficiency in full-stack architecture, Java systems, and front-end development(start\_span)(end\_span).
- Build professional growth skills, including debugging, repository management, and technical documentation(start\_span)(end\_span).

## 2.4 References

1. UCT Internship Project Guidelines & Deliverable Specifications(start\_span)(end\_span).
2. The IoT Academy: Java Standard Edition & J2SE Modules (Slides 1 to 6)(start\_span)(end\_span)(start\_span)(end\_span)(start\_span)(end\_span).
3. Oracle Java Documentation: `java.sql.PreparedStatement` and Java 8 Stream API(start\_span)(end\_span)(start\_span)(end\_span).
4. Full Stack Web Development Course Reader: Architecture, MEAN Stack, and Semantic HTML(start\_span)(end\_span)(start\_span)(end\_span)(start\_span)(end\_span).

2.5 Glossary & Acronyms

| Acronym | Full Term / Meaning              |
|---------|----------------------------------|
| UCT     | UniConverge Technologies Pvt Ltd |
| USC     | upSkill Campus                   |
| JDBC    | Java Database Connectivity       |
| OOP     | Object-Oriented Programming      |
| DOM     | Document Object Model            |
| OEE     | Overall Equipment Effectiveness  |
| SPA     | Single-Page Application          |
| REST    | Representational State Transfer  |

3. PROBLEM STATEMENT

In traditional automotive spare parts and transactional retail portals, users encounter fragmented inventories, high latency when filtering across complex specifications, and clunky checkout workflows. On the server side, unoptimized database queries and repetitive SQL parsing degrade system throughput. The assigned problem statement required engineering an end-to-end web system capable of delivering smooth front-end categorization and search alongside a structured data persistence workflow that eliminates repetitive server-side parsing bottlenecks.

4. EXISTING AND PROPOSED SOLUTION

Limitations of Existing Systems:

- Reliance on synchronous, full-page reloads that increase network payload and delay response times.
- Poorly categorized interfaces making part compatibility difficult to verify.

- Use of basic SQL statements without parameterization, leading to redundant query parsing in database software and increased vulnerability to injection attacks[span\_51](start\_span)[span\_51](end\_span).

### Proposed Solution & Value Addition:

- Built a responsive Single-Page Application (SPA) with instant keyword searching and category filtering (Brakes, Electrical, Fluids, Lighting)[span\_52](start\_span)[span\_52](end\_span).
- Implemented a persistent in-memory cart with live price calculations and dynamic DOM updates[span\_53](start\_span)[span\_53](end\_span).
- Designed backend transactional workflows utilizing pre-compiled PreparedStatement models to ensure single-parse query execution and high database efficiency[span\_54](start\_span)[span\_54](end\_span).

### Official Deliverable Submission Links:

- **Code Submission (GitHub Link):**  
<https://github.com/kausharwarish/upskillcampus/blob/main/BankingInformationSystem.java> [span\_55](start\_span)[span\_55](end\_span)
- **Report Submission (GitHub Link):**  
[https://github.com/kausharwarish/upskillcampus/blob/main/BankingInformationSystem\\_MD\\_KAUSHAR\\_WARISH\\_USC\\_UCT.pdf](https://github.com/kausharwarish/upskillcampus/blob/main/BankingInformationSystem_MD_KAUSHAR_WARISH_USC_UCT.pdf) [span\_56](start\_span)[span\_56](end\_span)

## 5. PROPOSED DESIGN & SYSTEM ARCHITECTURE

### 5.1 High-Level Diagram (HLD)

The system follows a three-tier architectural model: Presentation Layer (User Interface) ↔ Business Logic / Controller Layer ↔ Data Persistence / Storage Layer.

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[Client Browser: HTML5 / Tailwind CSS / Vanilla JS Engine]
    ↓ ↑ (Asynchronous Data Requests / JSON / REST)
[Application Controller & Business Logic Layer]
    ↓ ↑ (Driver Manager & Parameterized PreparedStatement)
[Relational / Document Database Storage (RDBMS / MongoDB)][span_57](start_span)[span_57](end_span)
[span_57](end_span)[span_58](start_span)[span_58](end_span)[span_59](start_span)
[span_59](end_span)[span_60](start_span)[span_60](end_span)

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### 5.2 Low-Level Diagram (LLD) & Data Flow

- **User Input Handling:** Catalog search queries and category selections trigger event listeners that filter the data model without reloading the document[span\_61](start\_span)[span\_61](end\_span).

- **Cart State Machine:** Handles addition, quantity tracking, removal, and real-time total computation before packaging order payloads(start\_span)[span\_62](end\_span).
- **Database Optimization:** Uses pre-compiled parameterized queries to minimize compilation overhead and manage database transactions safely(start\_span)[span\_63](end\_span)(start\_span)[span\_64](end\_span).

### 5.3 Interfaces & Integration

User interactions occur through clean modal interfaces and responsive grid structures(start\_span)[span\_65](end\_span). The data layer leverages standard Java JDBC API interfaces ( `Connection` , `PreparedStatement` , `ResultSet` ) to maintain robust connections across database environments(start\_span)[span\_66](end\_span)(start\_span)[span\_67](end\_span).

## 6. PERFORMANCE TESTING & CONSTRAINTS HANDLING

Industrial applications require identifying operational constraints and testing designs to ensure durability, low latency, and memory stability(start\_span)[span\_68](end\_span).

### 6.1 Test Plan & Constraints

| Constraint             | Risk / Impact                       | Mitigation in System Design                                                                          |
|------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------|
| UI Latency             | Sluggish re-rendering during search | Client-side array filtering directly in memory <span>(start_span)[span_69](end_span)</span> .        |
| Database Overhead      | Repeated query parsing slow-downs   | Pre-compiled <code>PreparedStatement</code> execution <span>(start_span)[span_70](end_span)</span> . |
| Form Submission Errors | Corrupted order data entries        | Client-side modal validation prior to payload dispatch.                                              |

### 6.2 Test Procedure

1. Simulated rapid catalog keyword inputs to monitor DOM rendering stability(start\_span)[span\_71](end\_span).
2. Tested multi-item cart additions, item removals, and total balance calculations across varied prices(start\_span)[span\_72](end\_span).
3. Executed repeated database queries under load to verify pre-compilation efficiency(start\_span)[span\_73](end\_span).

### 6.3 Performance Outcomes

- Catalog search filtering performed consistently with sub-50ms visual updates.

- Parameterized query execution eliminated redundant parsing overhead, ensuring safe and optimized transactions[span\_74](start\_span)[span\_74](end\_span).
- 100% of test order submissions passed validation checks without state conflicts.

## 7. MY LEARNINGS

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Over the course of the internship, I gained hands-on experience across multiple core areas of modern software development:

- **Object-Oriented Programming:** Mastered class design, constructors, memory management, inheritance, dynamic polymorphism, and encapsulation in Java[span\_75](start\_span)[span\_75](end\_span)[span\_76](start\_span)[span\_76](end\_span).
- **Modern Java 8 Features:** Applied Lambda Expressions, functional interfaces, the Stream API, and the `java.time` package for clean, readable code[span\_77](start\_span)[span\_77](end\_span).
- **Database Connectivity:** Gained deep understanding of JDBC architecture, Driver Manager workflows, and the performance advantages of `PreparedStatement` over simple statements[span\_78](start\_span)[span\_78](end\_span)[span\_79](start\_span)[span\_79](end\_span).
- **Full-Stack Architecture:** Learned how front-end markup and client scripts communicate with server controllers and databases[span\_80](start\_span)[span\_80](end\_span)[span\_81](start\_span)[span\_81](end\_span)[span\_82](start\_span)[span\_82](end\_span)[span\_83](start\_span)[span\_83](end\_span)[span\_84](start\_span)[span\_84](end\_span).
- **Professional Engineering Practices:** Built confidence using Git/GitHub for version control, repository structuring, and industrial-grade technical documentation[span\_85](start\_span)[span\_85](end\_span)[span\_86](start\_span)[span\_86](end\_span)[span\_87](start\_span)[span\_87](end\_span).

## 8. FUTURE WORK SCOPE

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- Integrate live third-party payment gateway webhooks (such as Stripe or Razorpay sandbox environments)[span\_88](start\_span)[span\_88](end\_